Problems - Week 3

- Circuit Variables
- Circuit Elements
- Simple Resistive Circuits

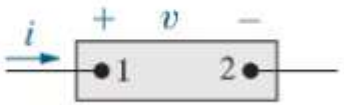
Circuit Variables

1.7 The current at the terminals of the element in Fig. 1.5 is

$$i = 0, \qquad t = 0;$$
$$i = 20e^{-5000t}\ \text{A}, \quad t \geq 0.$$

Calculate the total charge (in microcoulombs) entering the element at its left terminal.

Figure 1.5
An ideal basic circuit element.

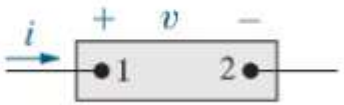


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Calculate the total charge (in microcoulombs) entering the element at its left terminal.

P 1.7 Remember from Eq. 1.2, current is the time rate of change of charge, or $i = \frac{dq}{dt}$ In this problem, we are given the current and asked to find the total charge. To do this, we must integrate Eq. 1.2 to find an expression for charge in terms of current:

$$q(t) = \int_0^t i(x)\,dx.$$

We are given the expression for current, i, which can be substituted into the above expression. To find the total charge, we let $t \to \infty$ in the integral. Thus we have

$$q_{\text{total}} = \int_0^\infty 20e^{-5000x}\,dx = \frac{20}{-5000}e^{-5000x}\Big|_0^\infty = \frac{20}{-5000}(e^{-\infty} - e^0)$$

$$= \frac{20}{-5000}(0 - 1) = \frac{20}{5000} = 0.004\ \text{C} = 4000\,\mu\text{C}.$$

Circuit Variables

1.8 There is no charge at the left terminal of the element in Fig. 1.5 for $t < 0$. At $t = 0$ a current of $25e^{-1000t}$ mA enters the left terminal.

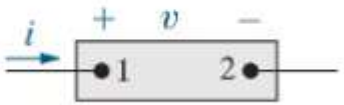
a. Derive the expression for the charge that accumulates at the left terminal for $t > 0$.

b. Find the total charge that accumulates at the left terminal.

c. If the current is stopped at $t = 1$ ms, how much charge has accumulated at the left terminal?

Figure 1.5

An ideal basic circuit element.



Circuit Variables

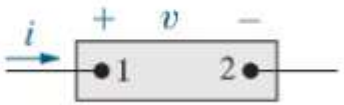
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c. If the current is stopped at $t = 1$ ms, how much charge has accur terminal?

Figure 1.5
An ideal basic circuit element.



[a] First we use Eq. 1.2 to relate current and charge:

$$i = \frac{dq}{dt} = 0.025e^{-1000t}.$$

Therefore, $dq = 0.025e^{-1000t}\,dt$.

To find the charge, we can integrate both sides of the last equation. Note that we substitute x for q on the left side of the integral, and y for t on the right side of the integral:

$$\int_{q(0)}^{q(t)} dx = 0.025 \int_0^t e^{-1000y}\,dy.$$

We solve the integral and make the substitutions for the limits of the integral:

$$q(t) - q(0) = 0.025\frac{e^{-1000y}}{-1000}\Bigg|_0^t = 25 \times 10^{-6}(1 - e^{-1000t}).$$

But $q(0) = 0$ by hypothesis, so

$$q(t) = 25(1 - e^{-1000t})\,\mu\text{C}.$$

Circuit Variables

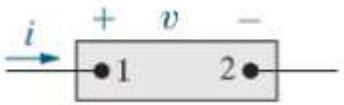
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An ideal basic circuit element.



$$q(t) = 25(1 - e^{-1000t})\ \mu\text{C}.$$

[b] As $t \to \infty$, $q_T = 25\ \mu\text{C}$.

Circuit Variables

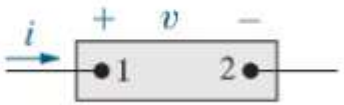
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Figure 1.5
An ideal basic circuit element.



[c] $q(1 \times 10^{-3}) = (25 \times 10^{-6})(1 - e^{(-1000)(0.001)}) = 15.8\,\mu\text{C}.$

Circuit Variables

1.12 The references for the voltage and current at the terminals of a circuit element are as shown in Fig. 1.6(c). The numerical values for v and i are 80V and −4A.

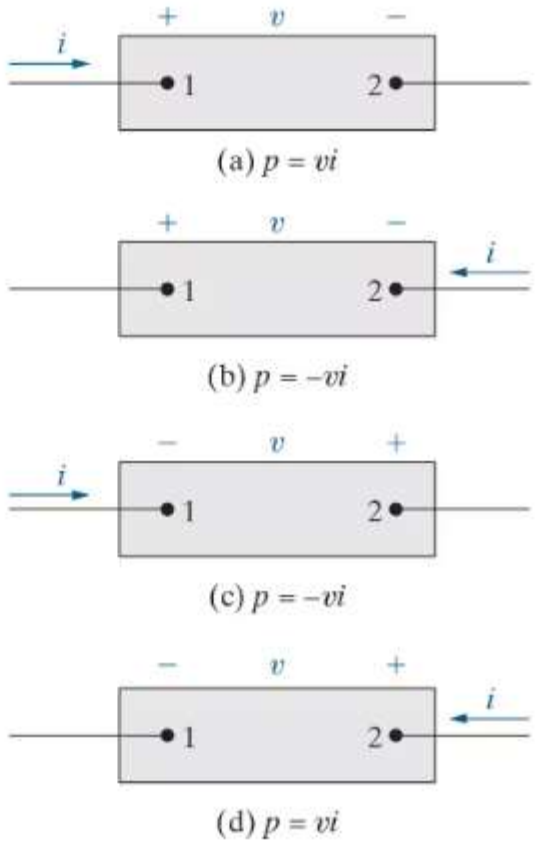
a. Calculate the power at the terminals and state whether the power is being absorbed or delivered by the element in the box.

b. Given that the current is due to electron flow, state whether the electrons are entering or leaving terminal 2.

c. Do the electrons gain or lose energy as they pass through the element in the box?

Figure 1.6

Polarity references and the expression for power.



Circuit Variables

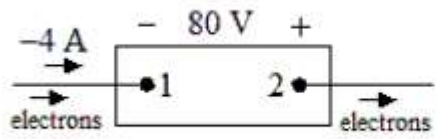
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[a]

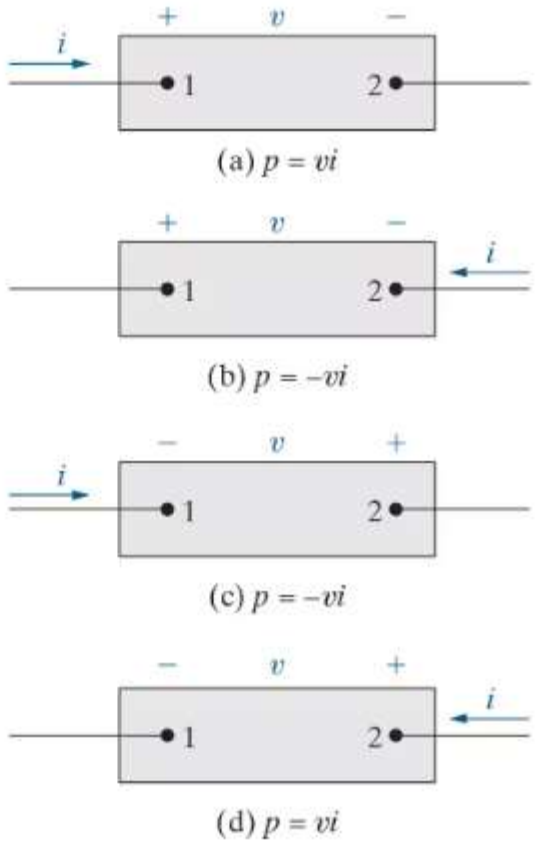


$p = vi = -(80)(-4) = 320$ W.
Power is being absorbed by the box.

[b] Leaving.

[c] Lose.

Figure 1.6
Polarity references and the expression for power.



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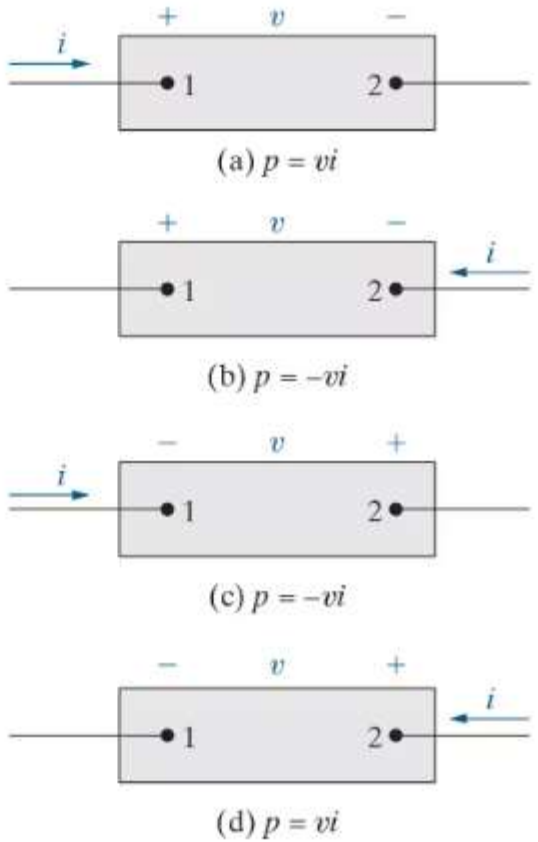
b. Given that the current is due to electron flow, state whether the electrons are entering or leaving terminal 2.

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1.13 Repeat Problem 1.12 with a current of 4 A.

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Polarity references and the expression for power.



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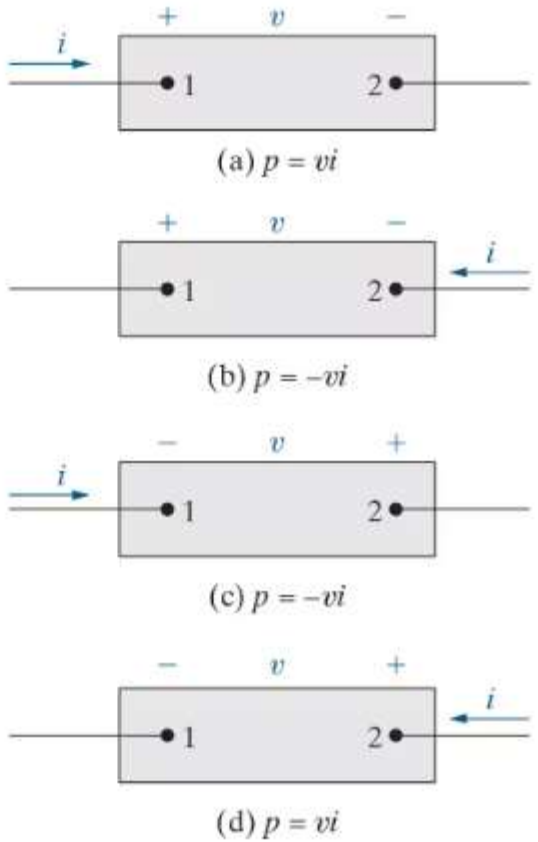
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1.13 Repeat Problem 1.12 with a current of 4 A.

Figure 1.6
Polarity references and the expression for power.



[a] $p = -vi = -(80)(4) = -320$ W, so power is being delivered by the box.

[b] Entering.

[c] Gain.

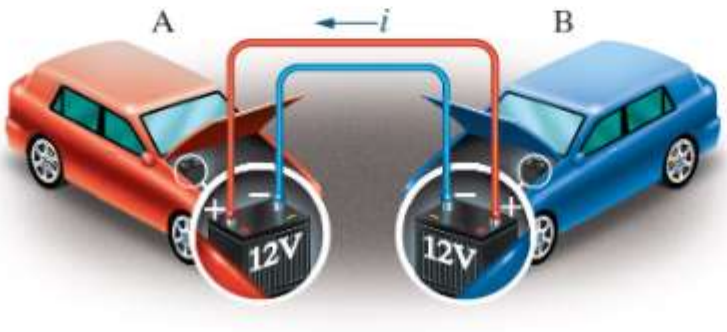
Circuit Variables

1.14 When a car has a dead battery, it can often be started by connecting the battery from another car across its terminals. The positive terminals are connected together as are the negative terminals. The connection is illustrated in Fig. P1.14. Assume the current i in Fig. P1.14 is measured and found to be 25 A.

a. Which car has the dead battery?

b. If this connection is maintained for 1 min, how much energy is transferred to the dead battery?

Figure P1.14



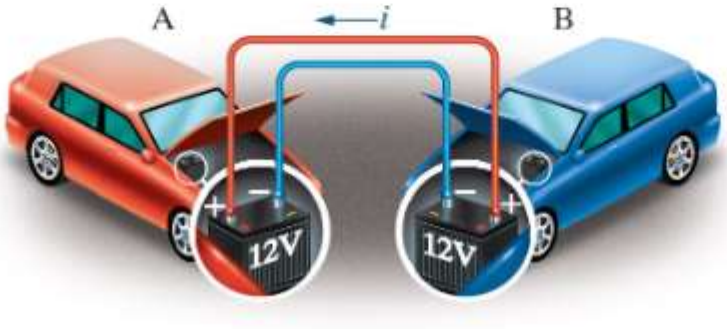
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a. Which car has the dead battery?

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Figure P1.14



[a] In Car A, the current i is in the direction of the voltage drop across the 12 V battery(the current i flows into the + terminal of the battery of Car A). Therefore using the passive sign convention,
$p = vi = (25)(12) = 300$ W.
Since the power is positive, the battery in Car A is absorbing power, so Car A must have the "dead" battery.

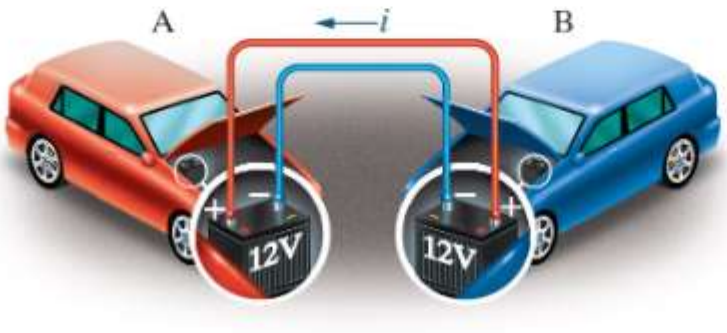
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[a] In Car A, the current i is in the direction of the voltage drop across the 12 V battery(the current i flows into the + terminal of the battery of Car A). Therefore using the passive sign convention,
$p = vi = (25)(12) = 300$ W.
Since the power is positive, the battery in Car A is absorbing power, so Car A must have the "dead" battery.

[b] $$w(t) = \int_0^t p\,dx; \qquad 1 \text{ min} = 1 \cdot \frac{60 \text{ s}}{1 \text{ min}} = 60 \text{ s};$$

$$w(60) = \int_0^{60} 300\,dx;$$

$$w = 300(60 - 0) = 300(60) = 18{,}000 \text{ J} = 18 \text{ kJ}.$$

Circuit Variables

1.15 Two electric circuits, represented by boxes A and B, are connected as shown in Fig. P1.15. The reference direction for the current i in the interconnection and the reference polarity for the voltage v across the interconnection are as shown in the figure. For each of the following sets of numerical values, calculate the power in the interconnection and state whether the power is flowing from A to B or vice versa.

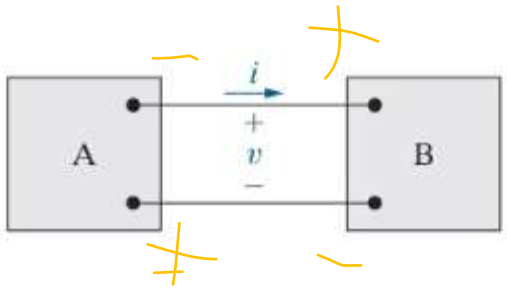
a. $i = 8$ A, $v = 40$ V

b. $i = -2$A, $v = -10$ V

c. $i = 2$A, $v = -50$ V

d. $i = -10$ A, $v = 20$ V

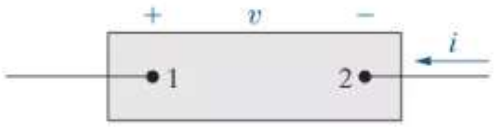
Figure P1.15



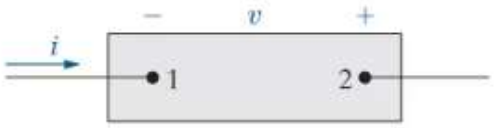
Circuit Variables



(a) $p = vi$



(b) $p = -vi$



(c) $p = -vi$



(d) $p = vi$

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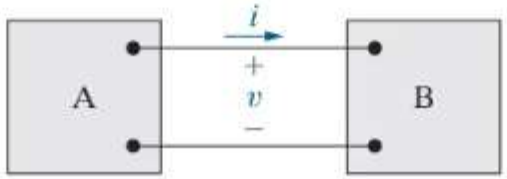
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b. $i = -2$A, $v = -10$ V

c. $i = 2$A, $v = -50$ V

d. $i = -10$ A, $v = 20$ V

Figure P1.15



Assume we are standing at box A looking toward box B. Use the passive sign convention to get $p = vi$, since the current i is owing into the + terminal of the voltage v. Now we just substitute the values for v and i into the equation for power. Remember that if the power is positive, B is absorbing power, so the power must be owing from A to B. If the power is negative, B is generating power so the power must be owing from B to A.

[a] $p = (40)(8) = 320$ W 320 W power is delivered from A to B;

[b] $p = (-10)(-2) = 20$ W 20 W power is delivered from A to B;

[c] $p = (-50)(2) = -100$ W 100 W power is delivered from B to A;

[d] $p = (20)(-10) = -200$ W 200 W power is delivered from B to A.

Circuit Variables

1.30 The numerical values for the currents and voltages in the circuit in Fig. P1.30 are given in Table P1.30. Find the total power developed in the circuit.

Figure P1.30

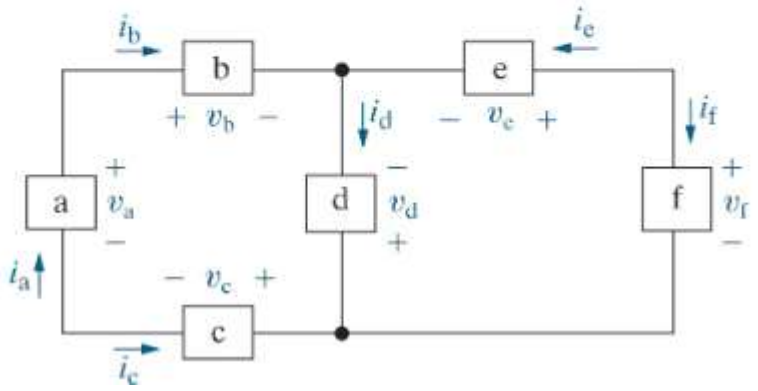


Table P1.30

Element	Voltage (V)	Current (mA)
a	40	−4
b	−24	−4
c	−16	4
d	−80	−1.5
e	40	2.5
f	120	−2.5

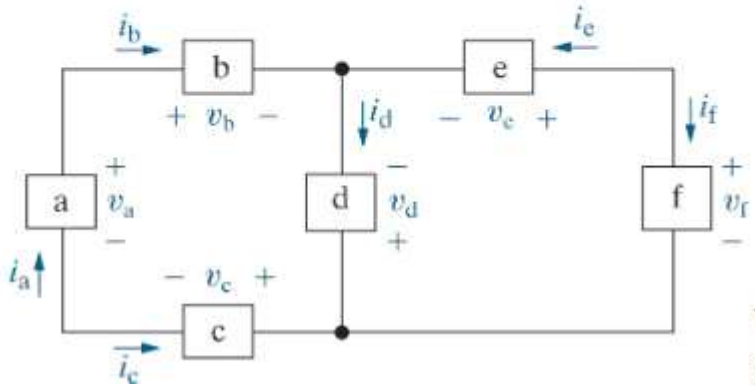
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1.30 The numerical values for the currents and voltages in the circuit in Fig. P1.30 are given in Table P1.30. Find the total power developed in the circuit.

Figure P1.30



We use the passive sign convention to determine whether the power equation is $p = vi$ or $p = -vi$ and substitute into the power equation the values for v and i, as shown below:

$$p_a = -v_a i_a = -(40)(-4 \times 10^{-3}) = 160 \text{ mW};$$
$$p_b = v_b i_b = (-24)(-4 \times 10^{-3}) = 96 \text{ mW};$$
$$p_c = -v_c i_c = -(-16)(4 \times 10^{-3}) = 64 \text{ mW};$$
$$p_d = -v_d i_d = -(-80)\left(-1.5 \times 10^{-3}\right) = -120 \text{ mW};$$
$$p_e = v_e i_e = (40)\left(2.5 \times 10^{-3}\right) = 100 \text{ mW};$$
$$p_f = v_f i_f = (120)\left(-2.5 \times 10^{-3}\right) = -300 \text{ mW}.$$

Circuit Variables

1.28 An industrial battery is charged over a period of several hours at a constant voltage of 120 V. Initially, the current is 20 mA and increases linearly to 30 mA in 10 ks. From 10 ks to 20 ks, the current is constant at 30 mA. From 20 ks to 30 ks the current decreases linearly to 10 mA. At 30 ks the power is disconnected from the battery.

a. Sketch the current from $t = 0$ to $t = 30$ ks.

b. Sketch the power delivered to the battery from $t = 0$ to $t = 30$ ks.

c. Using the sketch of the power, find the total energy delivered to the battery.

Circuit Variables

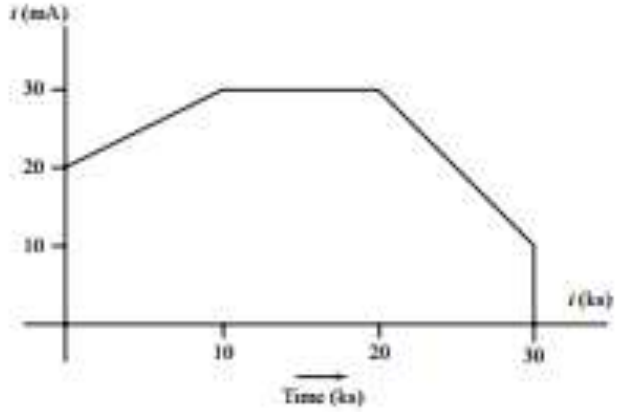
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[a]



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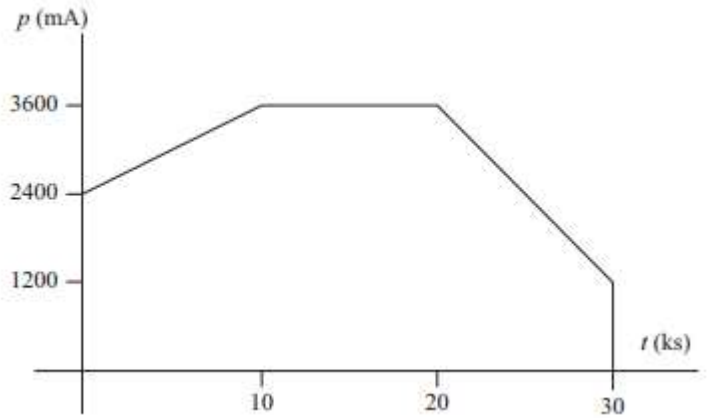
b. Sketch the power delivered to the battery from $t = 0$ to $t = 30$ ks.

c. Using the sketch of the power, find the total energy delivered to the battery.

[b]
$$\begin{aligned} i(t) &= 20 + 1.0 \times 10^{-3}t \text{ mA}, & 0 \le t \le 10 \text{ ks}; \\ i(t) &= 30 \text{ mA}, & 10 \text{ ks} \le t \le 20 \text{ ks}; \\ i(t) &= 70 - 2.0 \times 10^{-3}t \text{ mA}, & 20 \text{ ks} \le t \le 30 \text{ ks}; \\ i(t) &= 0, & t > 30 \text{ ks}. \end{aligned}$$

$p = vi = 120i$ so

$$\begin{aligned} p(t) &= 2400 + 0.12\text{t mW}, & 0 \le t \le 10 \text{ ks}; \\ p(t) &= 3600 \text{ mW}, & 10 \text{ ks} \le t \le 20 \text{ ks}; \\ p(t) &= 8400 - 0.24\text{t mW}, & 20 \text{ ks} \le t \le 30 \text{ ks}; \\ p(t) &= 0, & t > 30 \text{ ks}. \end{aligned}$$



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a. Sketch the current from $t = 0$ to $t = 30$ ks.

b. Sketch the power delivered to the battery from $t = 0$ to $t = 30$ ks.

c. Using the sketch of the power, find the total energy delivered to the battery.

[c] To find the energy, calculate the area under the plot of the power:

$$w\,(10\text{ ks}) = \frac{1}{2}\,(10{,}000)\,[2.4 + 3.6] = 25\text{ kJ};$$

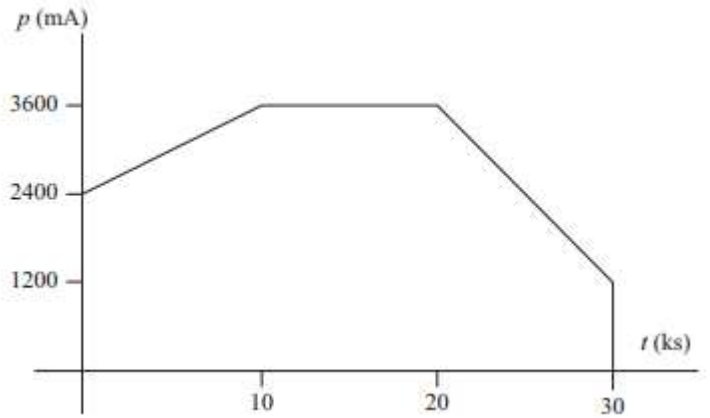
$$w\,(20\text{ ks}) = [10{,}000]\,[3.6] = 36\text{ kJ};$$

$$w\,(10\text{ ks}) = \frac{1}{2}\,(10{,}000)\,[3.6 + 1.2] = 24\text{ kJ};$$

[b]
$$i(t) = 20 + 1.0 \times 10^{-3}t\text{ mA}, \quad 0 \le t \le 10\text{ ks};$$
$$i(t) = 30\text{ mA}, \quad 10\text{ ks} \le t \le 20\text{ ks};$$
$$i(t) = 70 - 2.0 \times 10^{-3}t\text{ mA}, \quad 20\text{ ks} \le t \le 30\text{ ks};$$
$$i(t) = 0, \quad t > 30\text{ ks}.$$

$p = vi = 120i$ so

$$p(t) = 2400 + 0.12t\text{ mW}, \quad 0 \le t \le 10\text{ ks};$$
$$p(t) = 3600\text{ mW}, \quad 10\text{ ks} \le t \le 20\text{ ks};$$
$$p(t) = 8400 - 0.24t\text{ mW}, \quad 20\text{ ks} \le t \le 30\text{ ks};$$
$$p(t) = 0, \quad t > 30\text{ ks}.$$



Circuit Elements

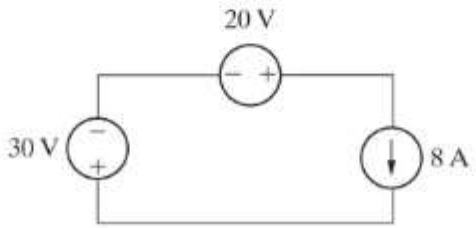
2.1 **a.** Is the interconnection of an ideal source in the circuit in Fig. P2.1 valid? Explain.

b. Identify which sources are developing power and which sources are absorbing power.

c. Verify that the total power developed in the circuit equals the total power absorbed.

d. Repeat (a)–(c), reversing the polarity of the 30 V source.

Figure P2.1



Circuit Elements

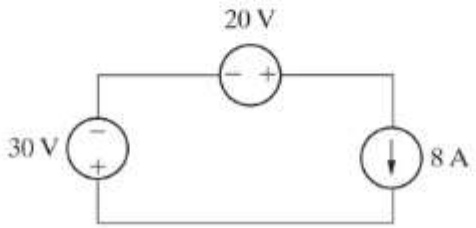
2.1 **a.** Is the interconnection of an ideal source in the circuit in Fig. P2.1 valid? Explain.

b. Identify which sources are developing power and which sources are absorbing power.

c. Verify that the total power developed in the circuit equals the total power absorbed.

d. Repeat (a)–(c), reversing the polarity of the 30 V source.

Figure P2.1



[a] Yes, independent voltage sources can carry the 8 A current required by the connection; independent current sources can support any voltage required by the connection, in this case 10 V, positive at the bottom.

[b] 30 V source: absorbing;

20 V source: developing (delivering);

8 A source: developing (delivering).

[c] $P_{30V} = (30)(8) = 240$ W

$P_{20V} = -(20)(8) = -160$ W

$P_{8V} = -(8)(10) = -80$ W

$\sum P_{abs} = \sum P_{del} = 240$ W

Circuit Elements

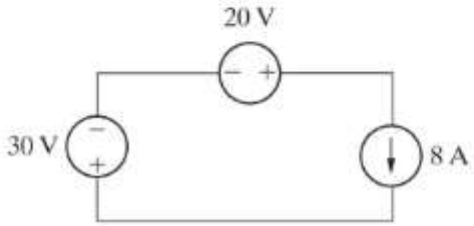
2.1 **a.** Is the interconnection of an ideal source in the circuit in Fig. P2.1 valid? Explain.

b. Identify which sources are developing power and which sources are absorbing power.

c. Verify that the total power developed in the circuit equals the total power absorbed.

d. Repeat (a)–(c), reversing the polarity of the 30 V source.

Figure P2.1



[d] The interconnection is valid, but in this circuit the voltage drop across the 8 A current source is 50 V, positive at the top; the 30 V source is developing (delivering), the 20 V source is developing (delivering), and the 8 A source is absorbing.

$$P_{30V} = -(30)(8) = -240 \text{ (dev/del)};$$

$$P_{20V} = -(20)(8) = -160 \text{ (dev/del)};$$

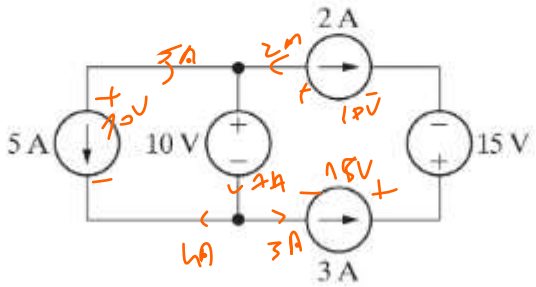
$$P_{8A} = (50)(8) = 400 \text{ (abs)}$$

$$\sum P_{abs} = \sum P_{del} = 400 \text{ W}$$

Circuit Elements

2.3 If the interconnection in Fig. P2.3 is valid, find the power developed by the voltage sources. If the interconnection is not valid, explain why.

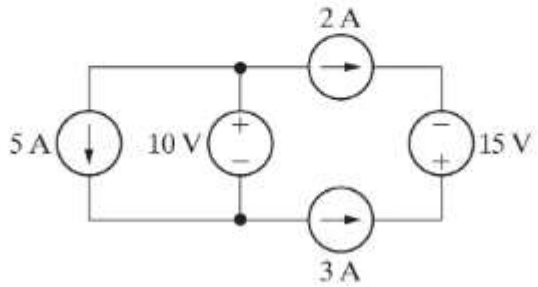
Figure P2.3



Circuit Elements

2.3 If the interconnection in Fig. P2.3 is valid, find the power developed by the voltage sources. If the interconnection is not valid, explain why.

Figure P2.3



P 2.3 The interconnection is not valid. Note that the 2 A and 3 A sources are both connected in the same branch of the circuit. A valid interconnection would require these two current sources to supply the same current in the same direction, which they do not.

Circuit Elements

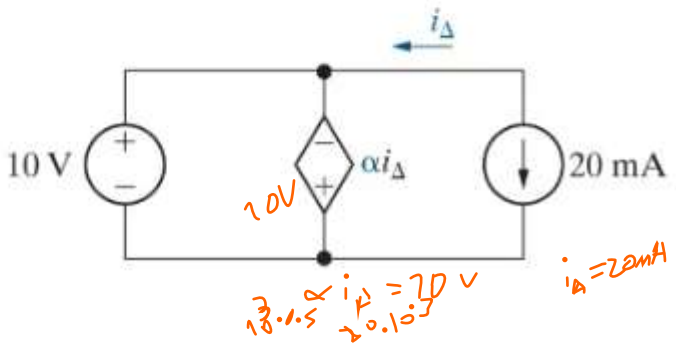
2.6 Consider the interconnection shown in Fig. P2.6.

a. What value of α is required to make this a valid interconnection?

b. For this value of α, find the power associated with the current source

c. Is the current source supplying or absorbing power?

Figure P2.6



Circuit Elements

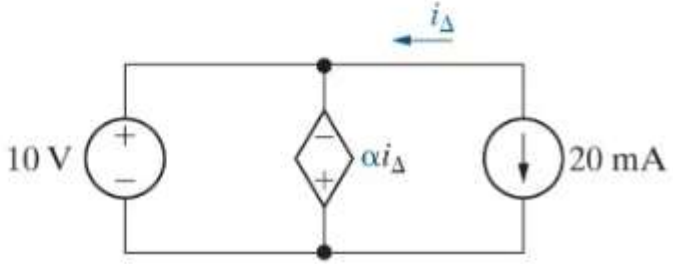
2.6 Consider the interconnection shown in Fig. P2.6.

a. What value of α is required to make this a valid interconnection?

b. For this value of α, find the power associated with the current source

c. Is the current source supplying or absorbing power?

Figure P2.6



[a] The voltage drop from the top node to the bottom node in this circuit must be the same for every path from the top to the bottom. Therefore, the voltages of the two voltage sources are equal:

$-\alpha i_\Delta = 10.$

Also, the current i_Δ is in the same branch as the 20 mA current source, but in the opposite direction, so

$i_\Delta = -0.020\text{A}.$

Substituting,

$$-\alpha\,(-0.020) = 10 \rightarrow \alpha = \frac{10}{0.02} = 500.$$

The interconnection is valid if $\alpha = 500$ V/A.

Circuit Elements

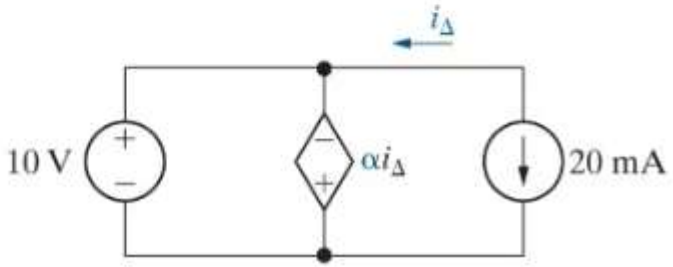
2.6 Consider the interconnection shown in Fig. P2.6.

a. What value of α is required to make this a valid interconnection?

b. For this value of α, find the power associated with the current source

c. Is the current source supplying or absorbing power?

Figure P2.6



[a] The voltage drop from the top node to the bottom node in this circuit must be the same for every path from the top to the bottom. Therefore, the voltages of the two voltage sources are equal:

$-\alpha i_\Delta = 10.$

Also, the current i_Δ is in the same branch as the 20 mA current source, but in the opposite direction, so

$i_\Delta = -0.020\text{A}.$

Substituting,

$$-\alpha\,(-0.020) = 10 \rightarrow \alpha = \frac{10}{0.02} = 500.$$

The interconnection is valid if $\alpha = 500$ V/A.

[b] The voltage across the current source must equal the voltage across the 10 V source, since both are connected between the top and bottom nodes. Using the passive sign convention,

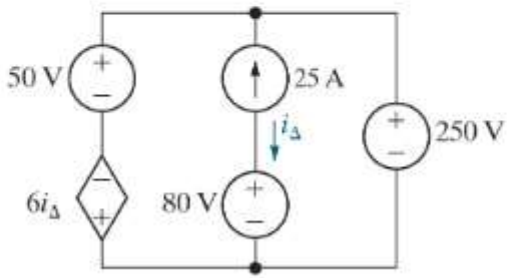
$p = vi = (10)(0.020) = 0.2\text{ W} = 200\text{ mW}.$

[c] Since the power is positive, the current source is absorbing power.

Circuit Elements

2.8 If the interconnection in Fig. P2.8 is valid, find the total power developed in the circuit. If the interconnection is not valid, explain why.

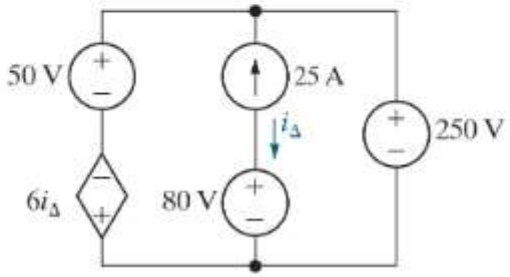
Figure P2.8



Circuit Elements

2.8 If the interconnection in Fig. P2.8 is valid, find the total power developed in the circuit. If the interconnection is not valid, explain why.

Figure P2.8



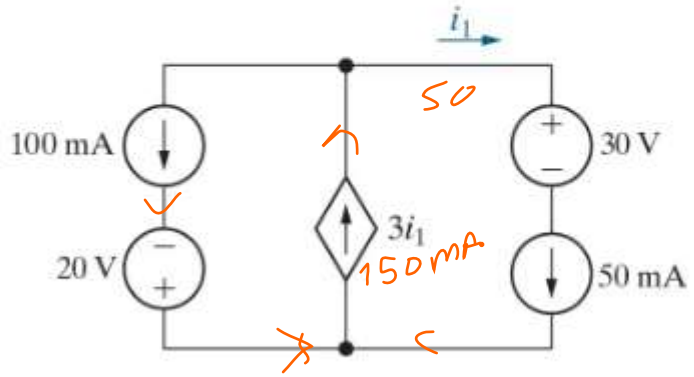
P 2.8 The interconnection is invalid. In the middle branch, the value of the current i_Δ must be -25 A, since the 25 A current source supplies current in this branch in the direction opposite the direction of the current i_Δ. Therefore, the voltage supplied by the dependent voltage source in the left hand branch is $6(-25) = -150$ V. This gives a voltage drop from the top terminal to the bottom terminal in the left hand branch of $50 - (-150) = 200$ V. But the voltage drop between these same terminals in the right hand branch is 250 V, due to the voltage source in that branch. Therefore, the interconnection is invalid.

Circuit Elements

2.10 **a.** Is the interconnection in Fig. P2.10 valid? Explain.

b. Can you find the total power developed in the circuit? Explain.

Figure P2.10

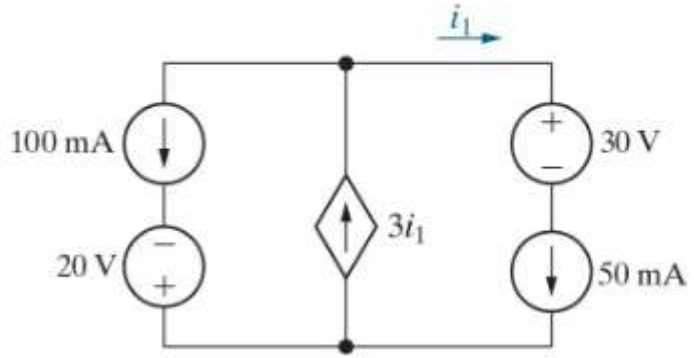


Circuit Elements

2.10 **a.** Is the interconnection in Fig. P2.10 valid? Explain.

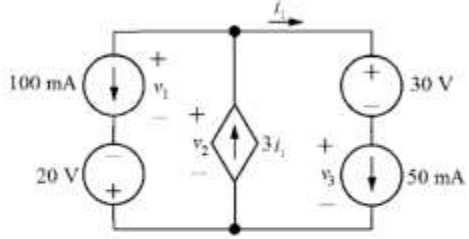
b. Can you find the total power developed in the circuit? Explain.

Figure P2.10



P 2.10 **[a]** Yes, each of the voltage sources can carry the current required by the interconnection, and each of the current sources can carry the voltage drop required by the interconnection. (Note that $i_1 = 50$ mA.)

[b] No, because the voltage drop between the top terminal and the bottom terminal cannot be determined. For example, define v_1, v_2, and v_3 as shown:



The voltage drop across the left branch, the center branch, and the right branch must be the same, since these branches are connected at the same two terminals. This requires that

$$v_1 - 20 = v_2 = v_3 + 30.$$

But this equation has three unknown voltages, so the individual voltages cannot be determined, and thus the power of the sources cannot be determined.

Circuit Elements

2.17 Given the circuit shown in Fig. P 2.17, find

a. the value of i_a,

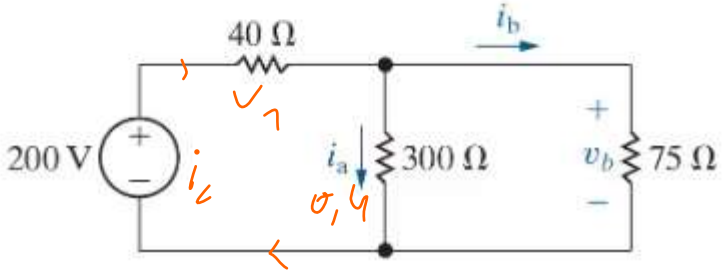
b. the value of i_b,

c. the value of v_b,

d. the power dissipated in each resistor,

e. the power delivered by the 200 V source.

Figure P2.17



Circuit Elements

2.17 Given the circuit shown in Fig. P 2.17, find

a. the value of i_a,

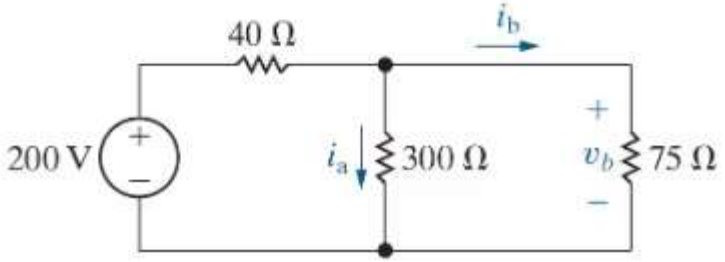
b. the value of i_b,

c. the value of v_b,

d. the power dissipated in each resistor,

e. the power delivered by the 200 V source.

Figure P2.17



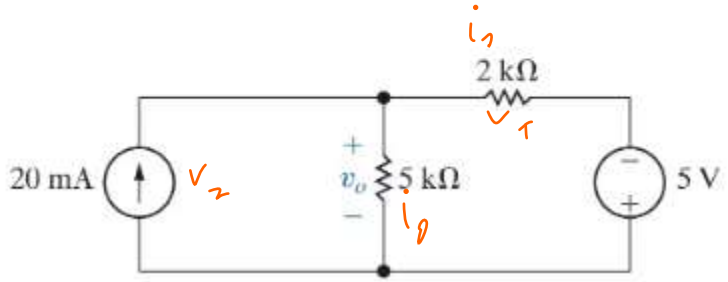
Circuit Elements

2.18 Consider the circuit shown in Fig. P 2.18.

a. Find v_o using Kirchoff's laws and Ohm's law.

b. Test the solution for v_o by verifying that the total power supplied equals the total power absorbed.

Figure P2.18



[a] KCL at the top node: $0.02 = i_1 + i_2$;

KVL around the right loop: $-v_o + v_2 - 5 = 0$.
Use Ohm's law to write the resistor voltages in the previous equation in terms of the resistor currents:

$$-5000i_1 + 2000i_2 - 5 = 0 \quad \rightarrow \quad -5000i_1 + 2000i_2 = 5.$$

Multiply the KCL equation by −2000 and add it to the KVL equation to eliminate i_2:

$$-2000(i_1+i_2)+(-5000i_1+2000i_2)=-2000(0.02)+5 \quad \rightarrow \quad -7000i_1=-35.$$

Solving,

$$i_1 = \frac{-35}{-7000} = 0.005 = 5 \text{ mA}.$$

Therefore,

$$v_o = Ri_1 = (5000)\,(0.005) = 25 \text{ V}.$$

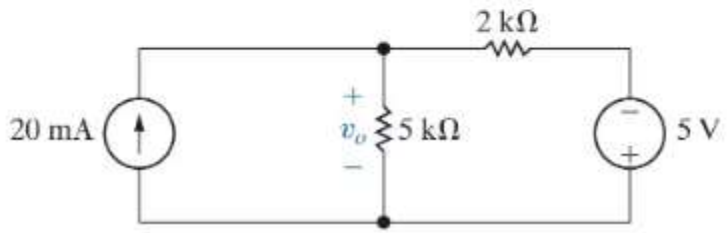
Circuit Elements

2.18 Consider the circuit shown in Fig. P 2.18.

a. Find υ_o using Kirchoff's laws and Ohm's law.

b. Test the solution for υ_o by verifying that the total power supplied equals the total power absorbed.

Figure P2.18



[a] KCL at the top node: $0.02 = i_1 + i_2$;

KVL around the right loop: $-v_o + v_2 - 5 = 0$.
Use Ohm's law to write the resistor voltages in the previous equation in terms of the resistor currents:

$$-5000i_1 + 2000i_2 - 5 = 0 \quad \rightarrow \quad -5000i_1 + 2000i_2 = 5.$$

Multiply the KCL equation by −2000 and add it to the KVL equation to eliminate i_2:

$$-2000(i_1+i_2)+(-5000i_1+2000i_2)=-2000(0.02)+5 \quad \rightarrow \quad -7000i_1=-35.$$

Solving,

$$i_1 = \frac{-35}{-7000} = 0.005 = 5 \text{ mA}.$$

Therefore,

$$v_o = Ri_1 = (5000)\,(0.005) = 25 \text{ V}.$$

[b] $p_{20\text{mA}} = -(0.02)\,v_o = -(0.02)\,(25) = -0.5$ W;

$i_2 = 0.02 - i_1 = 0.02 - 0.005 = 0.015$ A;

$p_{5\text{V}} = -(5)\,i_2 = -(5)\,(0.015) = -0.075$ W;

$p_{5\text{k}} = 5000i_1^2 = 5000(0.005)^2 = 0.125$ W;

$p_{2\text{k}} = 2000i_2^2 = 2000(0.015)^2 = 0.45$ W;

$p_{\text{total}} = p_{20\text{mA}} + p_{5\text{V}} + p_{5\text{k}} + p_{2\text{k}} = -0.5 - 0.075 + 0.125 + 0.45 = 0.$

Thus, the power in the circuit balances.

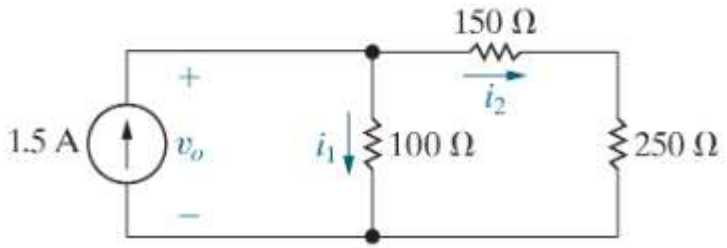
Circuit Elements

2.19 **a.** Find the currents i_1 and i_2 in the circuit in Fig. P 2.19.

b. Find the voltage v_o.

c. Verify that the total power developed equals the total power dissipated.

Figure P2.19



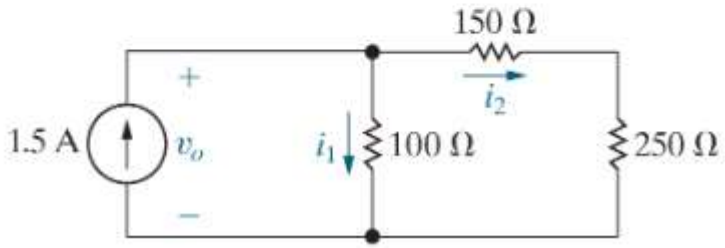
Circuit Elements

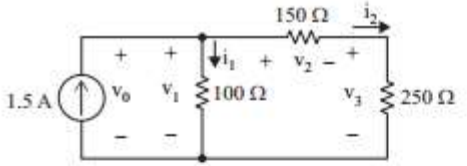
2.19 **a.** Find the currents i_1 and i_2 in the circuit in Fig. P 2.19.

b. Find the voltage v_o.

c. Verify that the total power developed equals the total power dissipated.

Figure P2.19





[a] Write a KCL equation at the top node:

$-1.5 + i_1 + i_2 = 0$ so $i_1 + i_2 = 1.5$

Write a KVL equation around the right loop:

$-v_1 + v_2 + v_3 = 0.$

From Ohm's law,

$v_1 = 100i_1, \quad v_2 = 150i_2, \quad v_3 = 250i_2.$

Substituting,

$-100i_1 + 150i_2 + 250i_2 = 0$ so $-100i_1 + 400i_2 = 0.$

Solving the two equations for i_1 and i_2 simultaneously,

$i_1 = 1.2$ A and $i_2 = 0.3$ A.

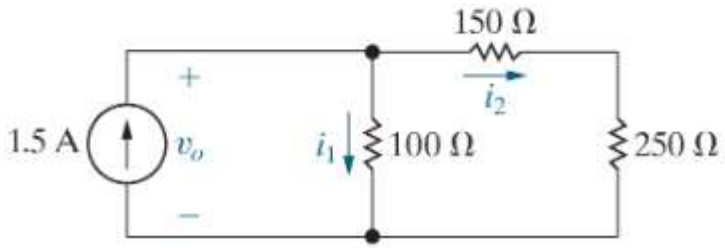
Circuit Elements

2.19 **a.** Find the currents i_1 and i_2 in the circuit in Fig. P 2.19.

b. Find the voltage v_o.

c. Verify that the total power developed equals the total power dissipated.

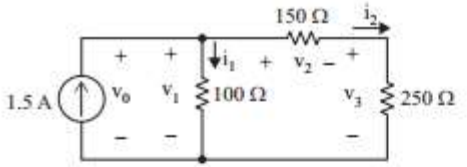
Figure P2.19



[b] Write a KVL equation clockwise around the left loop:

$-v_o + v_1 = 0$ but $v_1 = 100i_1 = 100\,(1.2) = 120$ V.

So $v_0 = v_1 = 120$ V.



[a] Write a KCL equation at the top node:

$-1.5 + i_1 + i_2 = 0$ so $i_1 + i_2 = 1.5$

Write a KVL equation around the right loop:

$-v_1 + v_2 + v_3 = 0.$

From Ohm's law,

$v_1 = 100i_1, \quad v_2 = 150i_2, \quad v_3 = 250i_2.$

Substituting,

$-100i_1 + 150i_2 + 250i_2 = 0$ so $-100i_1 + 400i_2 = 0.$

Solving the two equations for i_1 and i_2 simultaneously,

$i_1 = 1.2$ A and $i_2 = 0.3$ A.

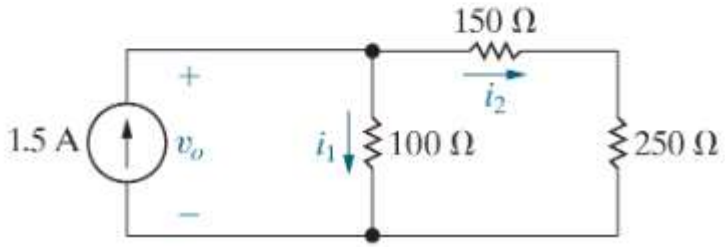
Circuit Elements

2.19 **a.** Find the currents i_1 and i_2 in the circuit in Fig. P 2.19.

b. Find the voltage v_o.

c. Verify that the total power developed equals the total power dissipated.

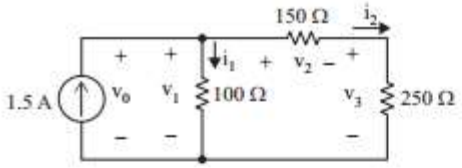
Figure P2.19



[b] Write a KVL equation clockwise around the left loop:

$-v_o + v_1 = 0$ but $v_1 = 100i_1 = 100\,(1.2) = 120$ V.

So $v_0 = v_1 = 120$ V.



[a] Write a KCL equation at the top node:

$-1.5 + i_1 + i_2 = 0$ so $i_1 + i_2 = 1.5$

Write a KVL equation around the right loop:

$-v_1 + v_2 + v_3 = 0.$

From Ohm's law,

$v_1 = 100i_1, \quad v_2 = 150i_2, \quad v_3 = 250i_2.$

Substituting,

$-100i_1 + 150i_2 + 250i_2 = 0$ so $-100i_1 + 400i_2 = 0.$

Solving the two equations for i_1 and i_2 simultaneously,

$i_1 = 1.2$ A and $i_2 = 0.3$ A.

[c] Calculate power using $p = vi$ for the source and $p = Ri^2$ for the resistors:

$p_{\text{source}} = -v_o\,(1.5) = -\,(120)\,(1.5) = -180$ W;

$p_{100\Omega} = 1.2^2(100) = 144$ W;

$p_{150\Omega} = 0.3^2(150) = 13.5$ W;

$p_{250\Omega} = 0.3^2(250) = 22.5$ W.

$\sum P_{\text{dev}} = 180$ W $\quad \sum P_{\text{abs}} = 144 + 13.5 + 22.5 = 180$ W.

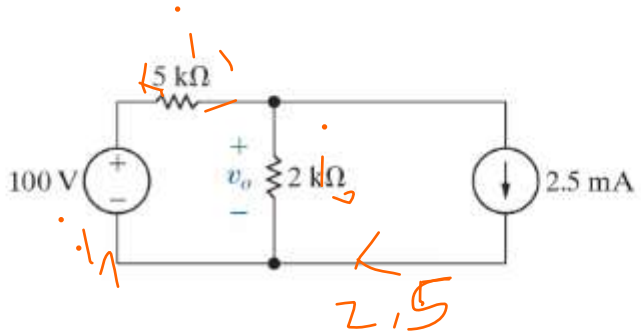
Circuit Elements

2.21 Consider the circuit shown in Fig. P2.21.

a. Find v_o using Kirchoff's laws and Ohm's law.

b. Test the solution for v_o by verifying that the total power supplied equals the total power absorbed.

Figure P2.21



[b] $i_1 = i_2 + 0.0025 = 0.0125 + 0.0025 = 0.015$ A;

$p_{100V} = -(100)i_1 = -(100)(0.015) = -1.5$ W;

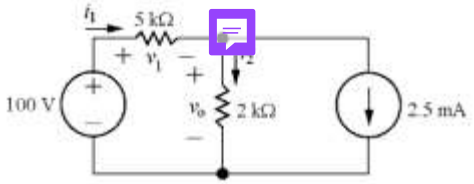
$p_{25mA} = (0.0025)v_o = (0.0025)(25) = 0.0625$ W;

$p_{5k} = 5000i_1^2 = 5000(0.015)^2 = 1.125$ W;

$p_{2k} = 2000i_2^2 = 2000(0.0125)^2 = 0.3125$ W;

$p_{total} = p_{25mA} + p_{100V} + p_{5k} + p_{2k} = 0.0625 - 1.5 + 1.125 + 0.3125 = 0.$

Thus the power in the circuit balances.



[a] KCL at the top node: $i_1 = i_2 + 0.0025;$

KVL around the left loop: $-100 + v_1 + v_o = 0.$

Use Ohm's law to write the resistor voltages in the previous equation in terms of the resistor currents:

$$-100 + 5000i_1 + 2000i_2 = 0 \quad \rightarrow \quad 5000i_1 + 2000i_2 = 100.$$

Use the KCL equation for i_1 to eliminate i_1 in the KVL equation:

$$5000(i_2 + 0.0025) + 2000i_2 = 100 \quad \rightarrow \quad 7000i_2 = 87.5.$$

Solving,

$$i_2 = \frac{87.5}{7000} = 0.0125 = 12.5 \text{ mA}.$$

Therefore,

$$v_o = Ri_2 = (2000)(0.0125) = 25 \text{ V}.$$

Circuit Elements

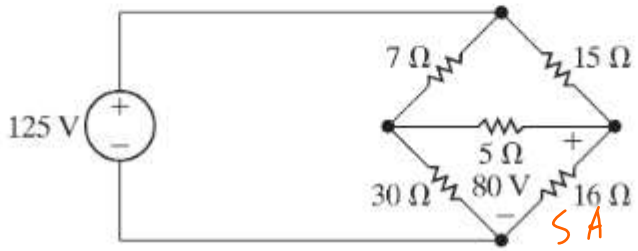
2.22 The voltage across the 16 Ω resistor in the circuit in Fig. P2.22 is 80 V, positive at the upper terminal.

a. Find the power dissipated in each resistor.

b. Find the power supplied by the 125 V ideal voltage source.

c. Verify that the power supplied equals the total power dissipated.

Figure P2.22



Circuit Elements

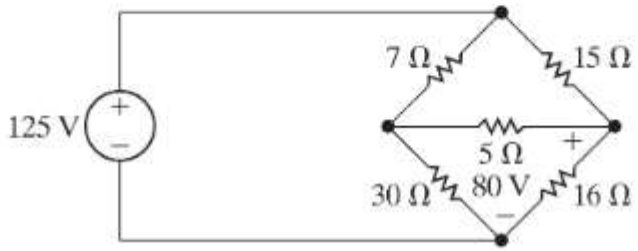
2.22 The voltage across the 16 Ω resistor in the circuit in Fig. P2.22 is 80 V, positive at the upper terminal.

a. Find the power dissipated in each resistor.

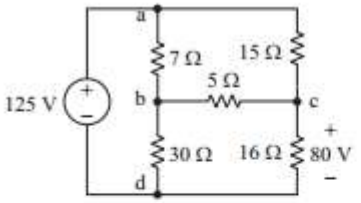
b. Find the power supplied by the 125 V ideal voltage source.

c. Verify that the power supplied equals the total power dissipated.

Figure P2.22



[a]



$i_{cd} = 80/16 = 5\text{A}$

$v_{ac} = 125 - 80 = 45$ so $i_{ac} = 45/15 = 3$ A;

$i_{ac} + i_{bc} = i_{cd}$ so $i_{bc} = 5 - 3 = 2$ A;

$v_{ab} = 15i_{ac} - 5i_{bc} = 15\,(3) - 5\,(2) = 35\text{V}$ so $i_{ab} = 35/7 = 5$ A;

$i_{bd} = i_{ab} - i_{bc} = 5 - 2 = 3$ A.

Calculate the power dissipated by the resistors using the equation $p_R = Ri_R^2$:

$p_{7\Omega} = (7)(5)^2 = 175\text{W}$ $p_{30\Omega} = (30)(3)^2 = 270$ W;

$p_{15\Omega} = (15)(3)^2 = 135\text{W}$ $p_{16\Omega} = (16)(5)^2 = 400$ W;

$p_{5\Omega} = (5)(2)^2 = 20\text{W}$.

Circuit Elements

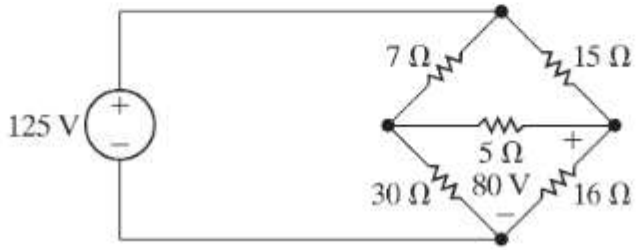
2.22 The voltage across the 16 Ω resistor in the circuit in Fig. P2.22 is 80 V, positive at the upper terminal.

a. Find the power dissipated in each resistor.

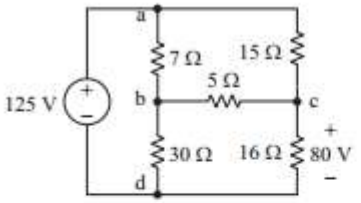
b. Find the power supplied by the 125 V ideal voltage source.

c. Verify that the power supplied equals the total power dissipated.

Figure P2.22



[a]



$i_{cd} = 80/16 = 5\text{A}$

$v_{ac} = 125 - 80 = 45 \quad \text{so} \quad i_{ac} = 45/15 = 3\ \text{A};$

$i_{ac} + i_{bc} = i_{cd} \quad \text{so} \quad i_{bc} = 5 - 3 = 2\ \text{A};$

$v_{ab} = 5i_{ac} - 5i_{bc} = 15\,(3) - 5\,(2) = 35\text{V} \quad \text{so} \quad i_{ab} = 35/7 = 5\ \text{A};$

$i_{bd} = i_{ab} - i_{bc} = 5 - 2 = 3\ \text{A}.$

Calculate the power dissipated by the resistors using the equation $p_R = Ri_R^2$:

$p_{7\Omega} = (7)(5)^2 = 175\text{W} \quad p_{30\Omega} = (30)(3)^2 = 270\ \text{W};$

$p_{15\Omega} = (15)(3)^2 = 135\text{W} \quad p_{16\Omega} = (16)(5)^2 = 400\ \text{W};$

$p_{5\Omega} = (5)(2)^2 = 20\text{W}.$

[b] Calculate the current through the voltage source:

$i_{125\text{V}} = -i_{ab} - i_{ac} = -5 - 3 = -8\text{A}.$

Now that we have both the voltage and the current for the source, we can calculate the power supplied by the source:

$p_g = 125\,(-8) = -1000\text{W} \quad \text{thus} \quad p_g\,(\text{supplied}) = 1000\text{W}.$

[c] $\sum P_{\text{dis}} = 175 + 270 + 135 + 400 + 20 = 1000\ \text{W}$

Therefore,

$\sum P_{\text{supp}} = \sum P_{\text{dis}}$

Circuit Elements

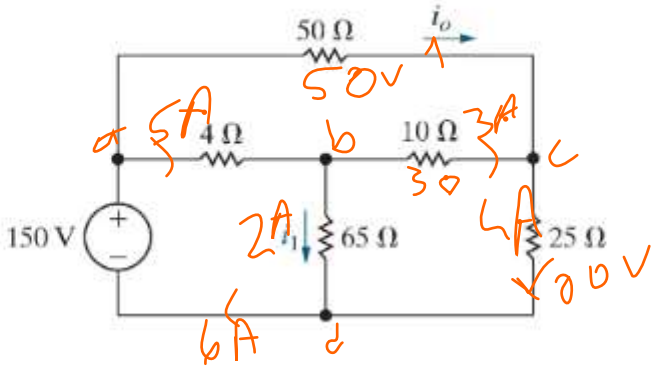
2.23 The current i_o in the circuit in Fig. P2.23 is 1 A.

a. Find i_1.

b. Find the power dissipated in each resistor.

c. Verify that the total power dissipated in the circuit equals the power developed by the 150 V source.

Figure P2.23



Circuit Elements

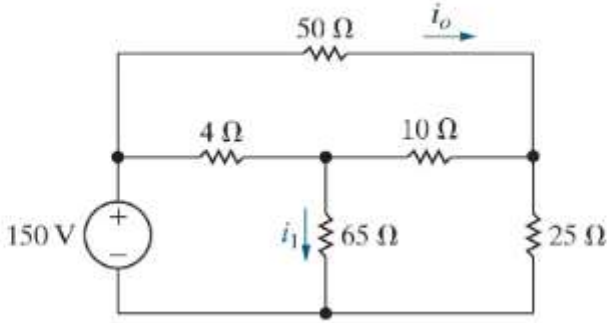
2.23 The current i_o in the circuit in Fig. P2.23 is 1 A.

a. Find i_1.

b. Find the power dissipated in each resistor.

c. Verify that the total power dissipated in the circuit equals the power developed by the 150 V source.

Figure P2.23



[b] $p_{4\Omega} = 5^2(4) = 100\ \text{W};$

$p_{50\Omega} = 1^2(50) = 50\ \text{W};$

$p_{65\Omega} = 2^2(65) = 260\ \text{W};$

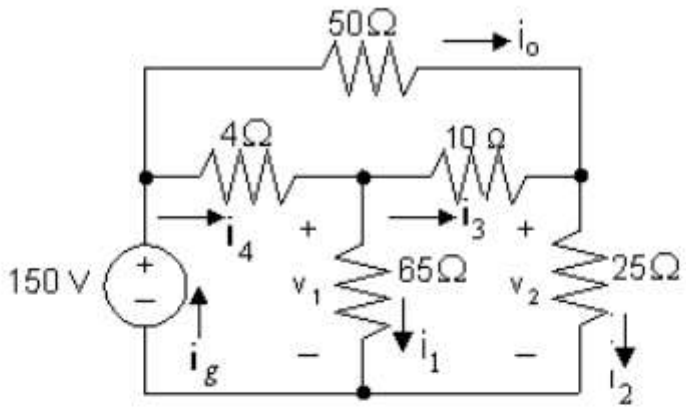
$p_{10\Omega} = 3^2(10) = 90\ \text{W};$

$p_{25\Omega} = 4^2(25) = 400\ \text{W}.$

[c] $\sum P_{\text{dis}} = 100 + 50 + 260 + 90 + 400 = 900\ \text{W};$

$P_{\text{dev}} = 150 i_g = 150(6) = 900\ \text{W}.$

[a]



$v_2 = 150 - 50(1) = 100\text{V};$

$i_2 = \dfrac{v_2}{25} = 4\text{A};$

$i_3 + 1 = i_2, \qquad i_3 = 4 - 1 = 3\text{A};$

$v_1 = 10 i_3 + 25 i_2 = 10(3) + 25(4) = 13($

$i_1 = \dfrac{v_1}{65} = \dfrac{130}{65} = 2\text{A}.$

Note also that

$i_4 = i_1 + i_3 = 2 + 3 = 5\,\text{A};$

$i_g = i_4 + i_o = 5 + 1 = 6\,\text{A}.$

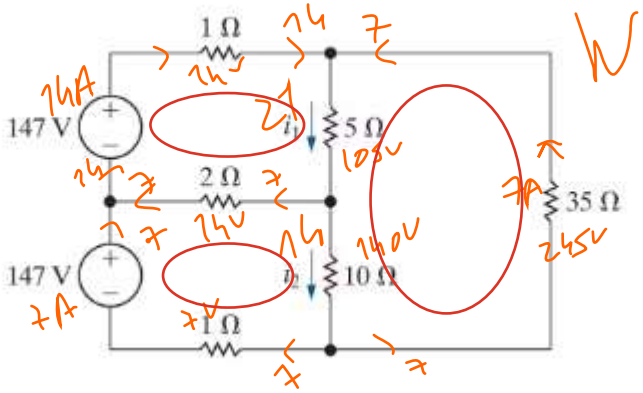
Circuit Elements

2.24 The currents i_1 and i_2 in the circuit in Fig. P2.24 are 21 A and 14 A, respectively.

a. Find the power supplied by each voltage source.

b. Show that the total power supplied equals the total power dissipated in the resistors.

Figure P2.24



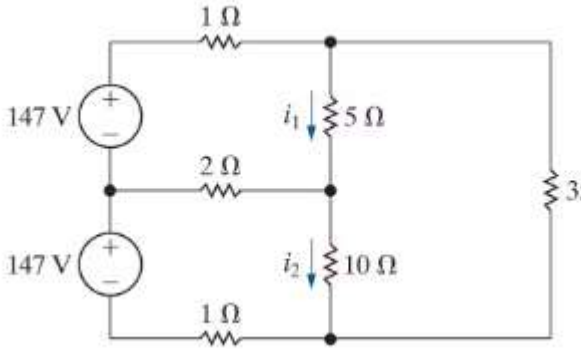
Circuit Elements

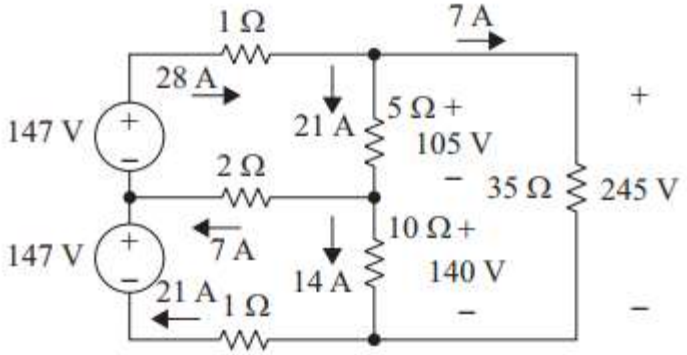
2.24 The currents i_1 and i_2 in the circuit in Fig. P2.24 are 21 A and 14 A, respectively.

a. Find the power supplied by each voltage source.

b. Show that the total power supplied equals the total power dissipated in the resistors.

Figure P2.24





$$p_{147}\,(\text{top}) = -\,(147)\,(28) = -4116 \text{ W};$$

$$p_{147}\,(\text{bottom}) = -\,(147)\,(21) = -3087 \text{ W}.$$

Therefore, the top source supplies 4116 W of power and the bottom source supplies 3087 W of power.

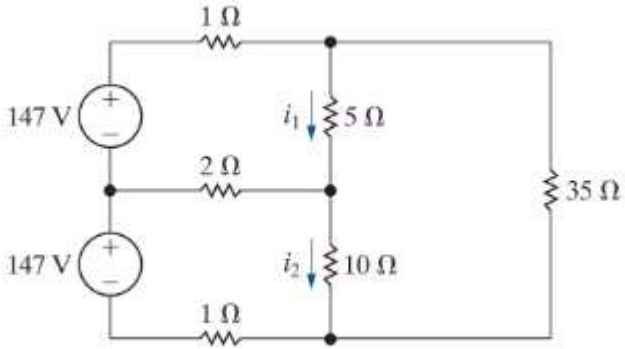
Circuit Elements

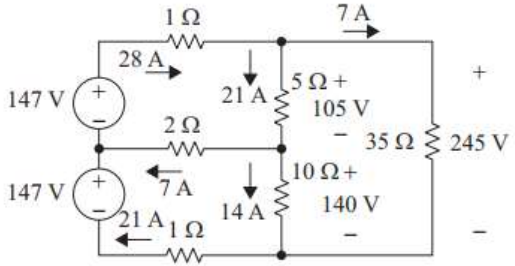
2.24 The currents i_1 and i_2 in the circuit in Fig. P2.24 are 21 A and 14 A, respectively.

a. Find the power supplied by each voltage source.

b. Show that the total power supplied equals the total power dissipated in the resistors.

Figure P2.24





$p_{147}\,(\text{top}) = -\,(147)\,(28) = -4116$ W;

$p_{147}\,(\text{bottom}) = -\,(147)\,(21) = -3087$ W.

Therefore, the top source supplies 4116 W of power and the bottom source supplies 3087 W of power.

[b] $\sum P_{\text{dis}} = (28)^2\,(1) + (7)^2\,(2) + (21)^2\,(1) + (21)^2\,(5) + (14)^2\,(10) + (7)^2\,(35)$

$= 784 + 98 + 441 + 2205 + 1960 + 1715 = 7203$ W.

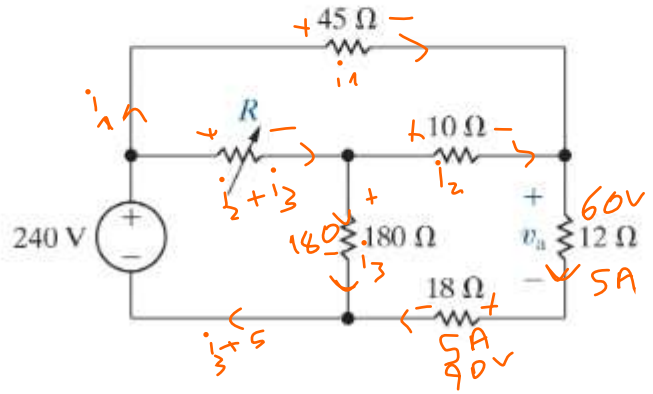
$\sum P_{\text{sup}} = 4116 + 3087 = 7203$ W.

Therefore, $\sum P_{\text{dis}} = \sum P_{\text{sup}} = 7203$ W.

Circuit Elements

2.26 The variable resistor R in the circuit in Fig. P2.26 is adjusted until v_a equals 60 V. Find the value of R.

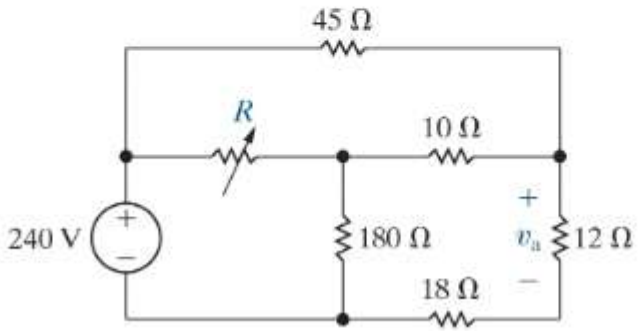
Figure P2.26

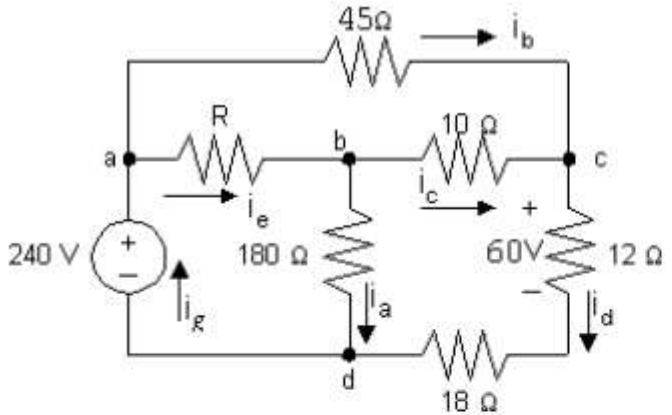


Circuit Elements

2.26 The variable resistor R in the circuit in Fig. P2.26 is adjusted until v_a equals 60 V. Find the value of R.

Figure P2.26



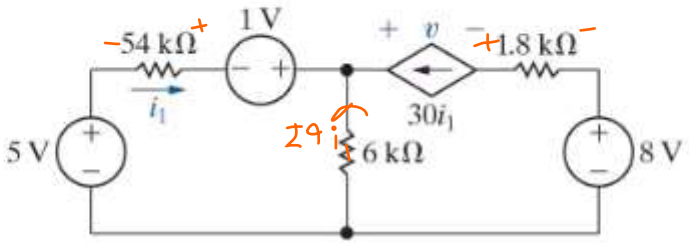


$i_d = 60/12 = 5\,\text{A}$; therefore, $v_{cd} = 60 + 18(5) = 150\,\text{V}$;
$-240 + v_{ac} + v_{cd} = 0$; therefore, $v_{ac} = 240 - 150 = 90\,\text{V}$;
$i_b = v_{ac}/45 = 90/45 = 2\,\text{A}$; therefore, $i_c = i_d - i_b = 5 - 2 = 3\,\text{A}$;
$v_{bd} = 10i_c + v_{cd} = 10(3) + 150 = 180\,\text{V}$;
therefore, $i_a = v_{bd}/180 = 180/180 = 1\,\text{A}$.
$i_e = i_a + i_c = 1 + 3 = 4\,\text{A}$;
$-240 + v_{ab} + v_{bd} = 0$; therefore, $v_{ab} = 240 - 180 = 60\,\text{V}$.
$R = v_{ab}/i_e = 60/4 = 15\,\Omega$.

Circuit Elements

2.33 For the circuit shown in Fig. P2.33, find (a) the current i_1 in microamperes, (b) the voltage v in volts, (c) the total power generated, and (d) the total power absorbed.

Figure P2.33

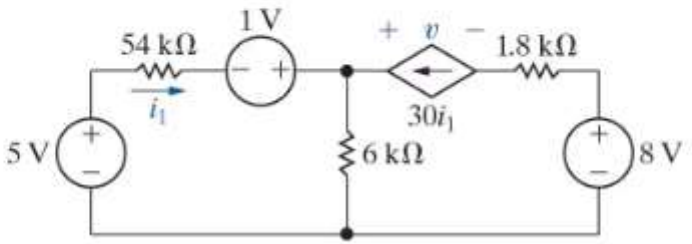


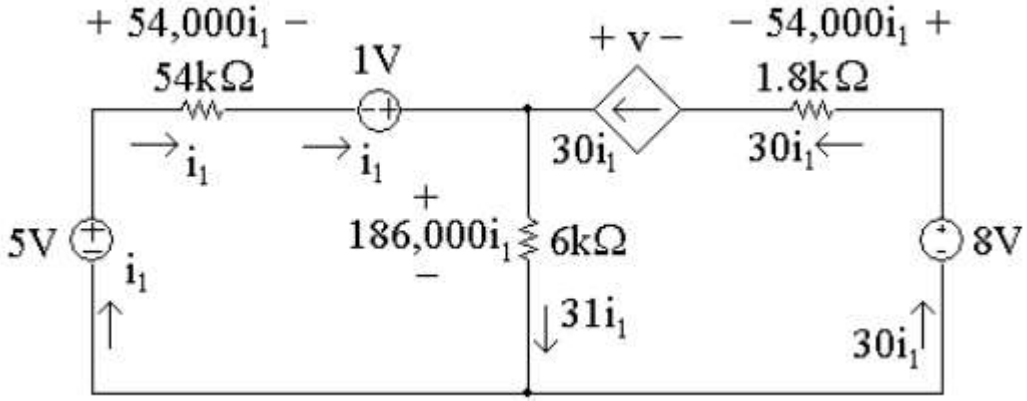
Circuit Elements

2.33 For the circuit shown in Fig. P2.33, find (a) the current i_1 in microamperes, (b) the voltage v in volts, (c) the total power generated, and (d) the total power absorbed.

$$i_1 + 30i_1 = i_{6k} = 31i_1.$$

Figure P2.33





$$-5\,\text{V} + 54{,}000i_1 - 1\,\text{V} + 186{,}000i_1 = 0.$$

Solving for i_1,

$$54{,}000i_1 + 186{,}000i_1 = 6\,\text{V} \qquad \text{so} \qquad 240{,}000i_1 = 6\,\text{V}.$$

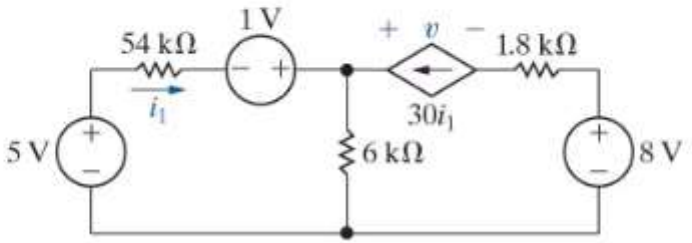
Thus,

$$i_1 = \frac{6}{240{,}000} = 25\,\mu\text{A}.$$

Circuit Elements

2.33 For the circuit shown in Fig. P2.33, find (a) the current i_1 in microamperes, (b) the voltage v in volts, (c) the total power generated, and (d) the total power absorbed.

Figure P2.33



$$+v - 54{,}000i_1 + 8\,\text{V} - 186{,}000i_1 = 0.$$

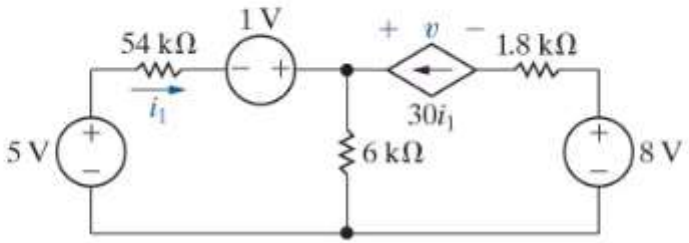
Thus,

$$v = 240{,}000i_1 - 8\,\text{V} = 240{,}000(25 \times 10^{-6}) - 8\,\text{V} = 6\,\text{V} - 8\,\text{V} = -2\,\text{V}.$$

Circuit Elements

2.33 For the circuit shown in Fig. P2.33, find (a) the current i_1 in microamperes, (b) the voltage v in volts, (c) the total power generated, and (d) the total power absorbed.

Figure P2.33



Element	Current (μA)	Voltage (V)	Power Equation	Power (μW)
5 V	25	5	$p = -vi$	−125
54 kΩ	25	1.35	$p = Ri^2$	33.75
1 V	25	1	$p = -vi$	−25
6 kΩ	775	4.65	$p = Ri^2$	3603.75
Dep. source	750	−2	$p = -vi$	1500
1.8 kΩ	750	1.35	$p = Ri^2$	1012.5
8 V	750	8	$p = -vi$	−6000

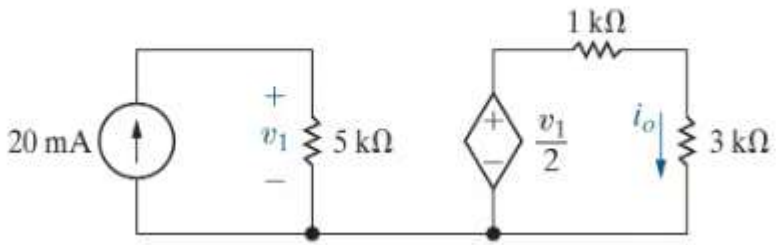
Circuit Elements

2.34 Consider the circuit shown in Fig. P2.34.

a. Find i_o.

b. Verify the value of i_o by showing that the power generated in the circuit equals the power absorbed in the circuit.

Figure P2.34



Circuit Elements

2.35 For the circuit in Fig. P2.35, current i_ϕ is 2 A. Calculate

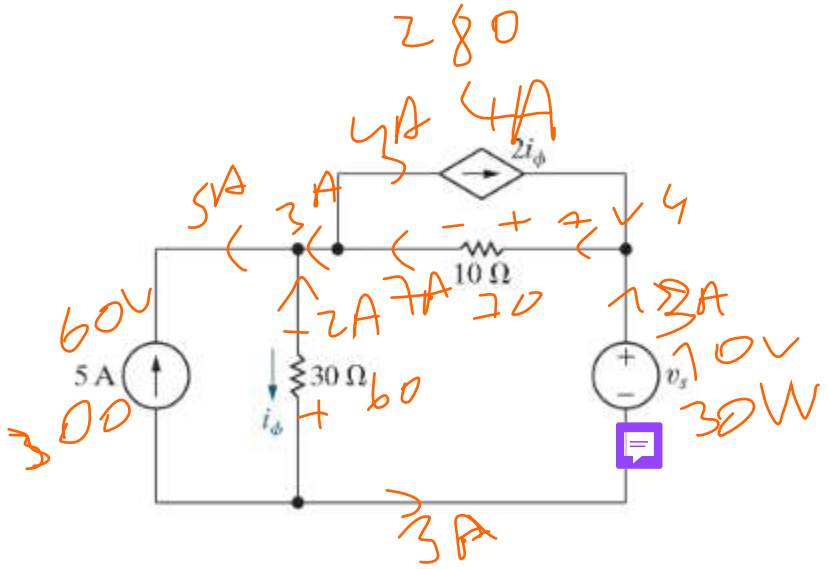
a. v_s,

b. the power absorbed by the independent voltage source,

c. the power delivered by the independent current source,

d. the power delivered by the controlled current source,**e.** the total power dissipated in the two resistors.

Figure P2.35



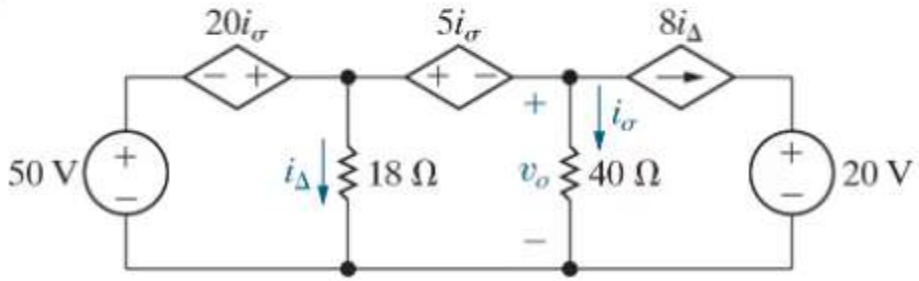
Circuit Elements

2.36 For the circuit shown in Fig. P2.36,

a. calculate i_Δ and i_o, and

b. show that the power developed equals the power absorbed.

Figure P2.36



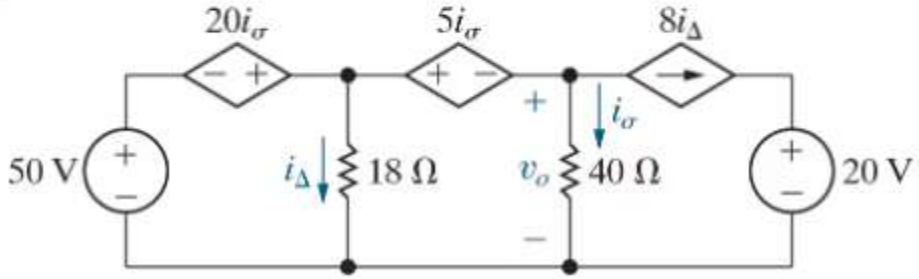
Circuit Element

2.36 For the circuit shown in Fig. P2.36,

a. calculate i_Δ and i_o, and

b. show that the power developed equals the power absorbed.

Figure P2.36



[a] $-50 - 20i_\sigma + 18i_\Delta = 0;$

$-18i_\Delta + 5i_\sigma + 40i_\sigma = 0, \quad \text{so} \quad 18i_\Delta = 45i_\sigma.$

Therefore, $\quad -50 - 20i_\sigma + 45i_\sigma = 0, \quad \text{so} \quad i_\sigma = 2\,\text{A}.$

$18i_\Delta = 45i_\sigma = 90;\ \text{so}\ i_\Delta = 5\,\text{A};$

$v_o = 40i_\sigma = 80\,\text{V}.$

[b] i_g = current out of the positive terminal of the 50 V source;
v_d = voltage drop across the $8i_\Delta$ source;

$i_g = i_\Delta + i_\sigma + 8i_\Delta = 9i_\Delta + i_\sigma = 47\,\text{A};$

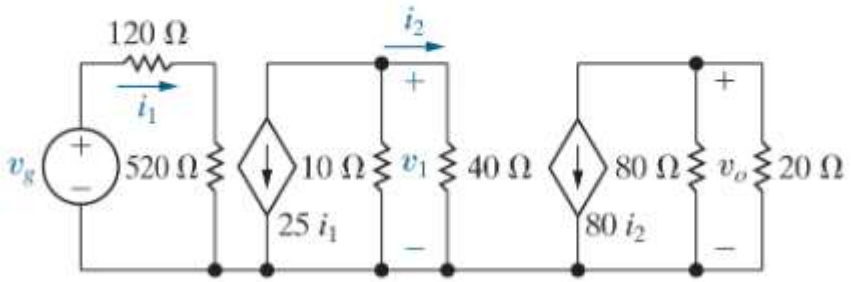
$v_d = 80 - 20 = 60\,\text{V}.$

$$\begin{aligned}
\sum P_\text{gen} &= 50i_g + 20i_\sigma i_g = 50(47) + 20(2)(47) = 4230\ \text{W};\\
\sum P_\text{diss} &= 18i_\Delta^2 + 5i_\sigma(i_g - i_\Delta) + 40i_\sigma^2 + 8i_\Delta v_d + 8i_\Delta(20)\\
&= (18)(25) + 10(47 - 5) + 4(40) + 40(60) + 40(20)\\
&= 4230\ \text{W; Therefore,}\\
\sum P_\text{gen} &= \sum P_\text{diss} = 4230\ \text{W}.
\end{aligned}$$

Circuit Elements

2.37 Find υ_1 and υ_g in the circuit shown in Fig. P2.37 when υ_o equals 10 V. (*Hint:* Start at the right end of the circuit and work back toward υ_g.)

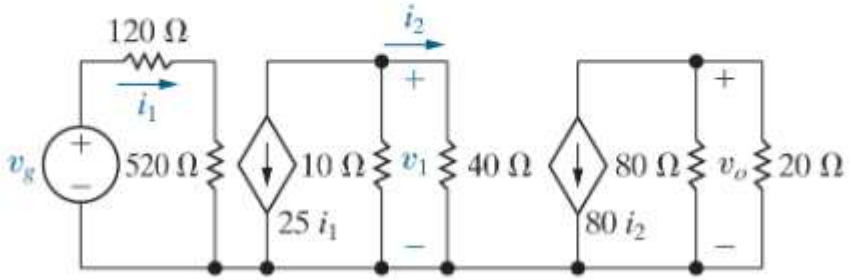
Figure P2.37



Circuit Elements

2.37 Find υ_1 and υ_g in the circuit shown in Fig. P2.37 when υ_o equals 10 V. (*Hint:* Start at the right end of the circuit and work back toward υ_g.)

Figure P2.37



P 2.37 $$80i_2 + \frac{10}{80} + \frac{10}{20} = 0; \qquad i_2 = -7.8125 \text{ mA};$$

$$\upsilon_1 = 40i_2 = -0.3125 \text{ V};$$

$$25i_1 + \frac{(-0.3125)}{10} + \left(-7.8125 \times 10^{-3}\right) = 0; \qquad i_1 = 1.5625 \text{ mA};$$

$$\upsilon_g = 120i_1 + 520i_1 = 640i_1.$$

Therefore, $\upsilon_g = 1$ V.

Simple Resistive Circuits

3.52 The bridge circuit shown in Fig. 3.28 is energized from a 24 V dc source. The bridge is balanced when $R_1 = 500\,\Omega$, $R_2 = 1000\,\Omega$, and $R_3 = 750\,\Omega$.

a. What is the value of R_x?

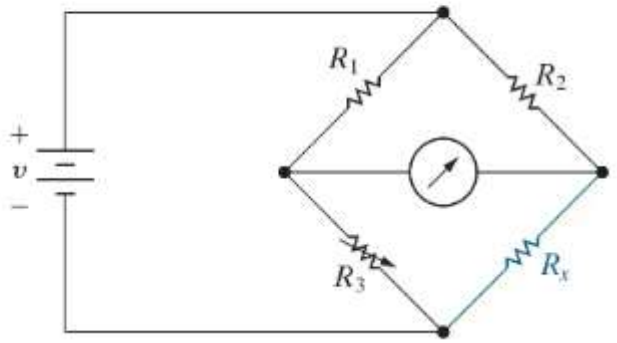
b. How much current (in milliamperes) does the dc source supply?

c. Which resistor in the circuit absorbs the most power? How much power does it absorb?

d. Which resistor absorbs the least power? How much power does it absorb?

Figure 3.28

The Wheatstone bridge circuit.



Simple Resistive Circuits

3.52 The bridge circuit shown in Fig. 3.28 is energized from a 24 V dc source. The bridge is balanced when $R_1 = 500\,\Omega$, $R_2 = 1000\,\Omega$, and $R_3 = 750\,\Omega$.

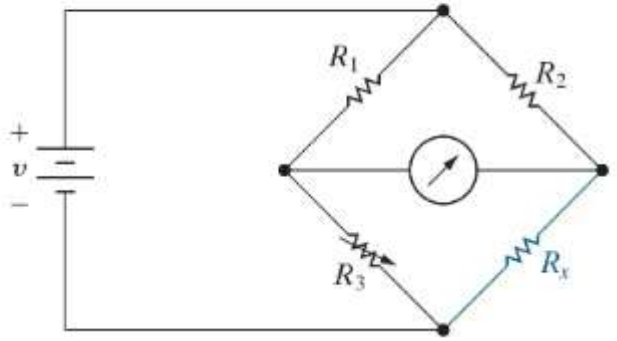
a. What is the value of R_x?

b. How much current (in milliamperes) does the dc source supply?

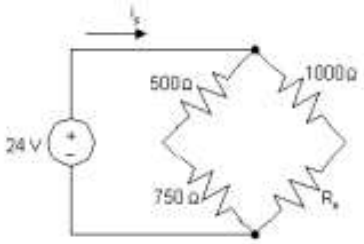
c. Which resistor in the circuit absorbs the most power? How much power does it absorb?

d. Which resistor absorbs the least power? How much power does it absorb?

Figure 3.28
The Wheatstone bridge circuit.



[a]



The condition for a balanced bridge is that the product of the opposite resistors must be equal:

$$(500)(R_x) = (1000)(750) \quad \text{so} \quad R_x = \frac{(1000)(750)}{500} = 1500\,\Omega.$$

$$i_s = \frac{24\text{V}}{500\,\Omega + 750\,\Omega} + \frac{24\,\text{V}}{1000\,\Omega + 1500\,\Omega} = 28.8\,\text{mA}.$$

[c] We can use Ohm's law to find the current in each branch:

$$i_{\text{left}} = \frac{24}{500 + 750} = 19.2\,\text{mA};$$

$$i_{\text{right}} = \frac{24}{1000 + 1500} = 9.6\,\text{mA}.$$

Now we can use the formula $p = Ri^2$ to find the power dissipated by each resistor:

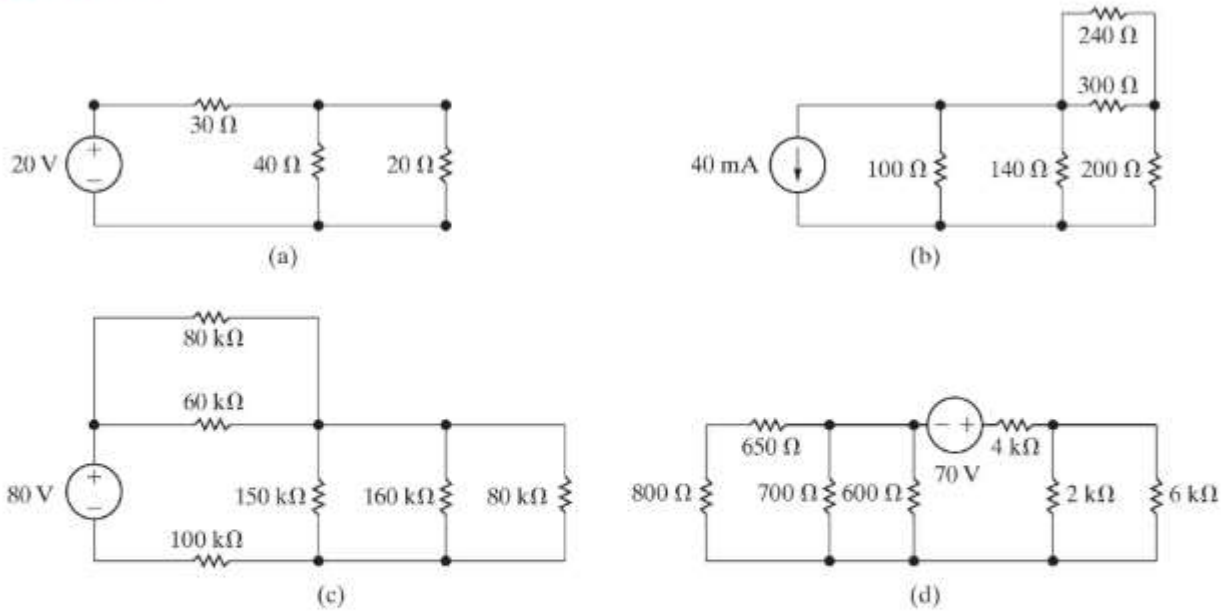
$$p_{500} = (500)(0.0192)^2 = 184.32\,\text{mW}; \qquad p_{750} = (750)(0.0192)^2 = 276.18\,\text{mW};$$

$$p_{1000} = (1000)(0.0096)^2 = 92.16\,\text{mW}; \qquad p_{1500} = (1500)(0.0096)^2 = 138.24\,\text{mW}.$$

Thus, the 750 Ω resistor absorbs the most power; it absorbs 276.48 mW of power.

Simple Resistive Circuits

Figure P3.2



3.3 For each of the circuits shown in Fig. P3.1:

a. Find the equivalent resistance seen by the source.

b. Find the power developed by the source.

Simple Resistive Circuits

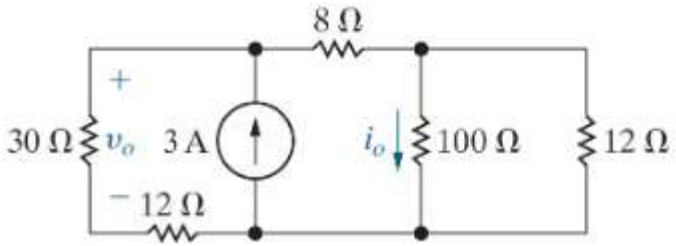
3.20 For the current divider circuit in Fig. P3.20 calculate

a. the current in the 8 Ω resistor,

b. the voltage across the 30 Ω resistor, and

c. the power dissipated in the 100 Ω resistor.

Figure P3.20



[a] $R_{eq} = [(100||12) + 8]\,||(30 + 12) = (10.71 + 8)||\,42 = 18.71||42 = 12.94\Omega$

$V_{3A} = (12.94)3 = 38.82V$

$$v_{8\Omega} = \frac{8}{8 + (12||100)}38.82 = \frac{8}{8 + 10.71}38.82 = 16.6\ \text{V};$$

$$i_{8\Omega} = \frac{16.6}{8} = 2.075A$$

[b] $$v_0 = \frac{30}{30\Omega + 12\Omega}38.82 = 27.73\ \text{V}.$$

[c] $v_{100\Omega} = 38.82 - 16.6 = 22.22\ \text{V};$

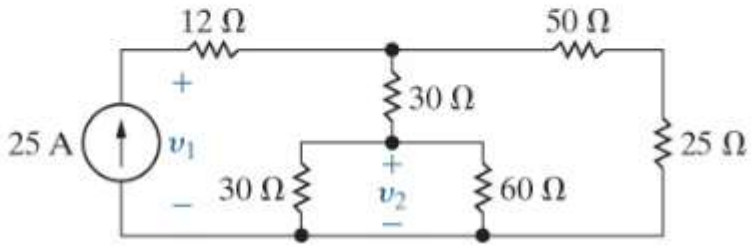
$$I_{100} = \frac{v_{100\Omega}}{100\Omega} = 0.22\ \text{A}.$$

$$P_1 = I^2R = (0.22)^2 \times 100 = 4.84\ \text{W}.$$

Simple Resistive Circuits

3.28 Find v_1 and v_2 in the circuit in Fig. P3.28 using voltage and/or current division.

Figure P3.28



$$60\|30 = 20\,\Omega;$$

$$i_{30\Omega} = \frac{50+25}{30+20+50+75}(25) = 15\,\text{A};$$

$$v_2 = (15)(20) = 300\,\text{V};$$

$$v_2 + 30i_{30} = 750\,\text{V};$$

$$v_1 - 12(25) = 750;$$

$$v_1 = 1050\,\text{V}.$$